

The robot is ready. The hospital isn't.

Robotic surgery made the procedure precise.
AI must make the episode around it coherent.

THE QUESTION

Can the institution move with the same precision as the machine?

ONE ROBOTIC CASE · 48 HOURS OUT

47:36

hours to incision

● Authorization	Evidence requested
● OR + robot	Block secured
● Scrub coverage	2-hour gap
● Bed capacity	1 staffed bed

Hospital economics no longer permit coordination waste.

2024

\$1.635T

U.S. hospital spending

2024

↗8.9%

one-year growth

AHA 2025

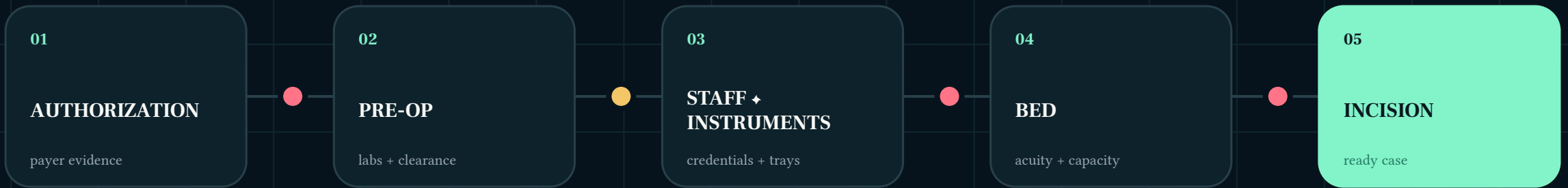
56%

of hospital costs are labor

Every unused minute, delayed bed-day, and failed handoff now compounds across the most expensive system in the economy.

No system owns the whole clock.

Every function sees its task. The risk accumulates between them.

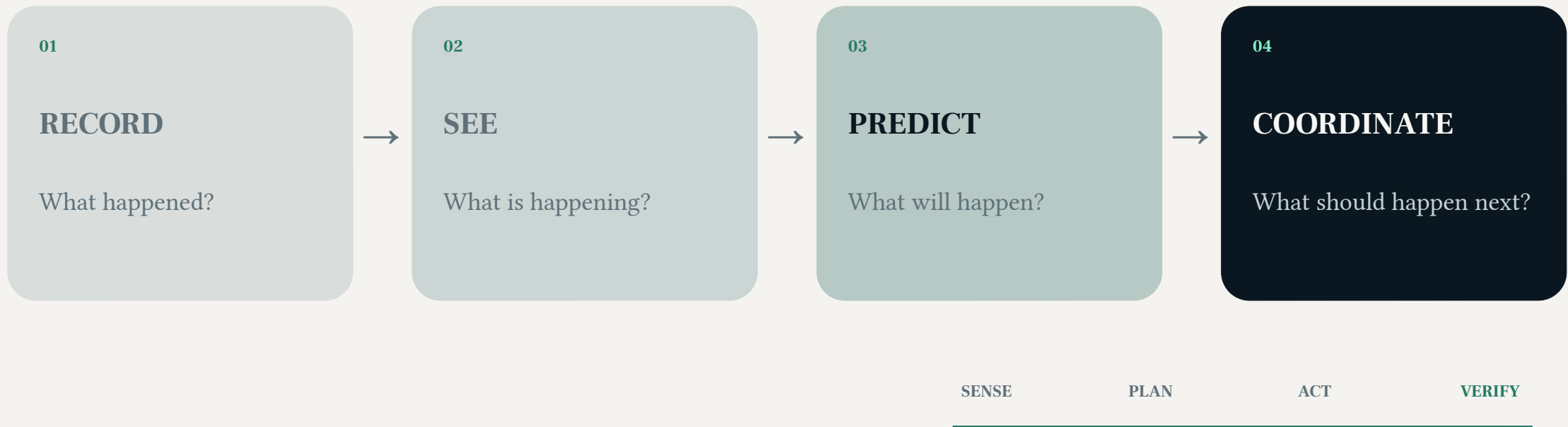


The bottleneck is not intelligence inside a system.

It is accountability across systems.

The next value layer is execution.

A dashboard predicts failure. A governed agent helps prevent it—and verifies what happened.



Six levers. One operating system.

The strategic mistake is funding six portfolios. The opportunity is connecting one system.



SHARED CONTEXT

One longitudinal view of intent, capacity, risk, and state

GOVERNED ACTION

Permissions, thresholds, named approvals, rollback, and stop conditions

CAUSAL SCORECARD

Access · flow · workforce · economics · trust

Each lever changes the unit of management.

DX

Clinical Diagnosis

Episode → continuously updated routing decision

LC

Longitudinal Care

Visits → continuously managed queue

FD

Digital Front Door

Channel → persistent relationship that resolves needs

TA

Task Automation

Task → governed outcome, executed and verified

RX

Robotics

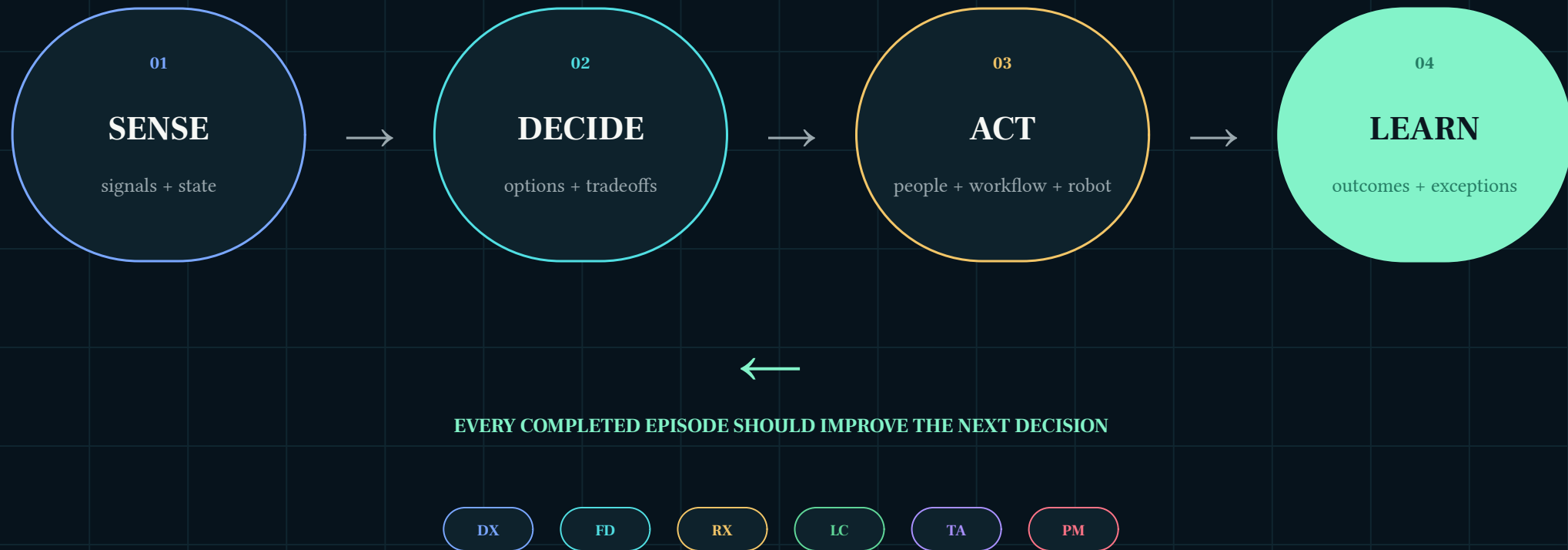
Precision instrument → distributed capacity platform

PM

Precision Medicine

Specialist report → learning operating model

Value compounds only when the loop closes.



Watch the constraint move.

RUN GUIDED DEMO

← BACK

DEPLOY ROBOTICS →

REPLAY CYCLE

EXIT STAGE

RESET

10 / 19
STATES | CONSTRAINT MOVES

● PATIENT FLOW
327 / 600
+29 vs baseline

SYSTEM CONSTRAINT
Operate

FLOW YIELD
54.5%
+4.8 pts vs baseline

MEDIAN JOURNEY
11d
4d faster

PEAK QUEUE
97
Operate stage

ADMIN TOUCHES
20
4 fewer vs baseline

04 · CONSTRAINT MOVES

The robot waits for the hospital.

Room readiness, instrument flow, and turnover now govern usable surgical capacity.

● Same demand trace · metrics change only when the operating rule changes

Four handoffs can still strand one robotic case.

The robot and surgeon are booked. Ordinary coordination failures still threaten the episode.

● ACCESS

Authorization evidence requested

Payer API · 4 min

● READINESS

Pre-op lab still pending

LIS · 21 min

● CAPACITY

Credentialed scrub coverage gap

Workforce · live

● FLOW

Only one staffed critical-care bed

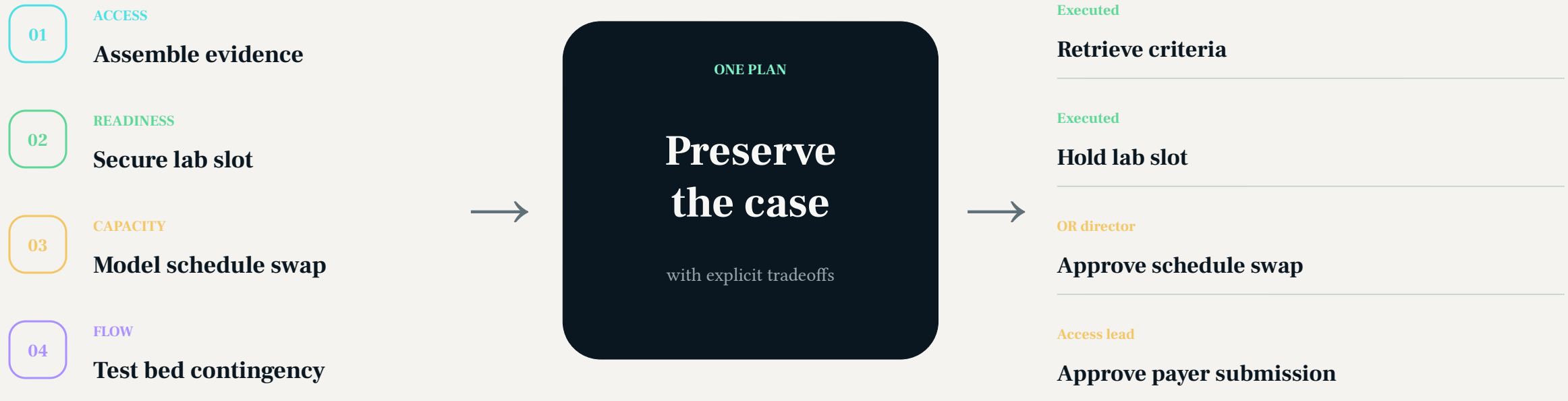
Bed command · live

47H 36M TO INCISION

Synthetic case · no patient data

Not a chat window. A governed decision system.

Goal ♦ policy ♦ permissions ♦ stop conditions → one constrained plan



Reversible work executes automatically. Consequential actions stop at named human approvals.

Same robot. More care.

No clinical judgment was delegated. Coordination latency was compressed around existing assets and staff.

23 → 7

manual touches

0 → 96

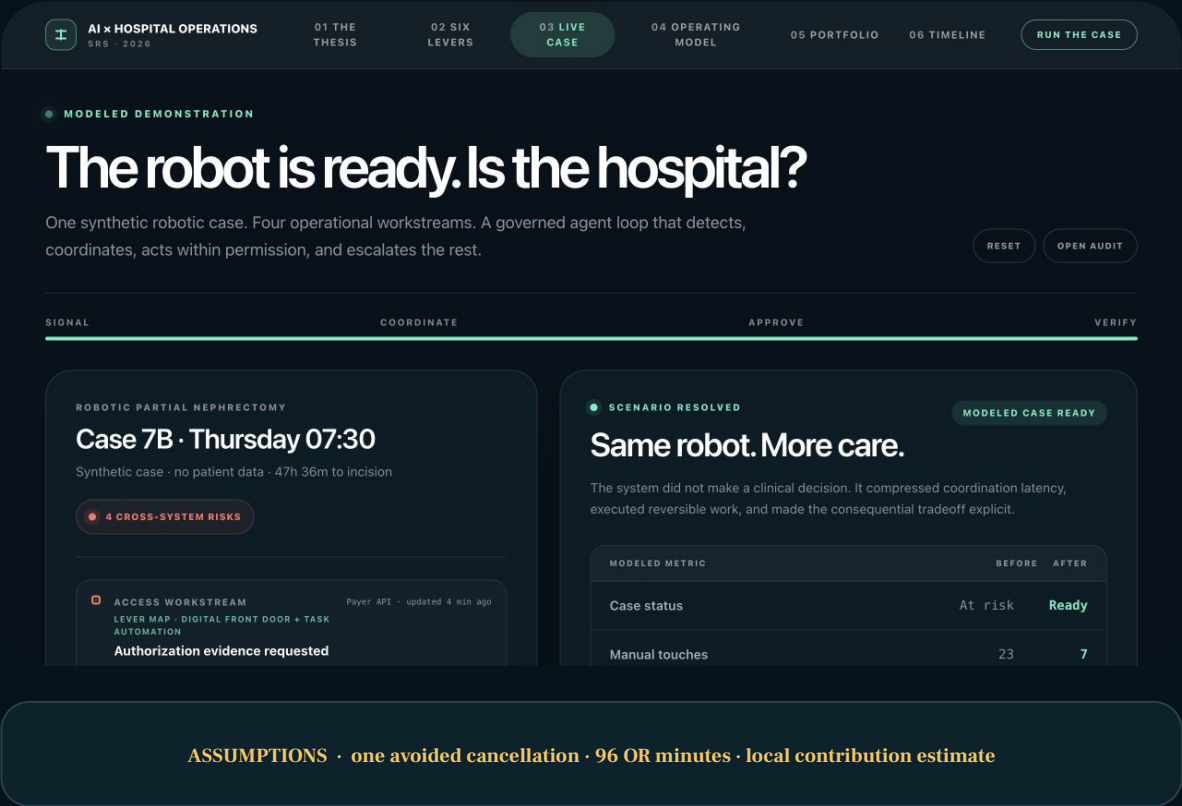
OR minutes recovered

6.5h → 1.8h

overtime exposure

\$0 → \$41K

modeled margin protected



Embedded AI is already moving operational outcomes.

Early implementations show the value of intelligence inside the workflow—not beside it.

HARTFORD HEALTHCARE

- **0.63**

days of length of stay

\$52–67M estimated annual benefit

OBSERVATIONAL IMPLEMENTATION

PENN ORACLE

884

fewer scheduling-error hours / year

5% more accurate OR scheduling

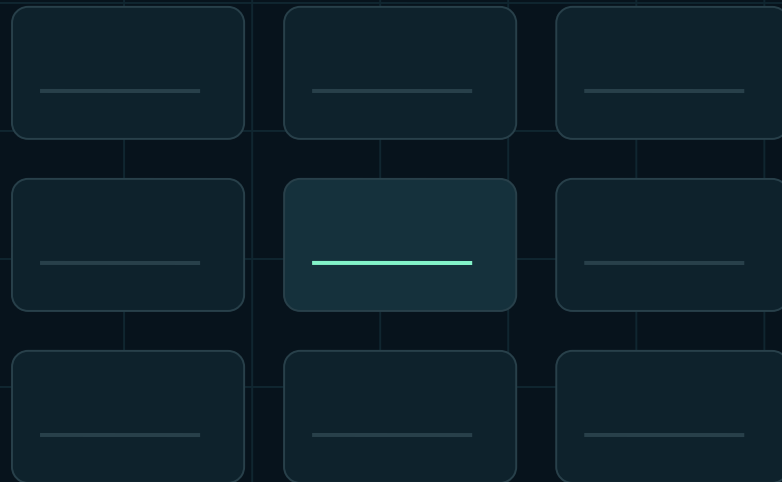
HEALTH-SYSTEM IMPLEMENTATION

These results support the mechanism. They do not yet prove a fully agentic hospital.

- THE WARNING

Visibility alone does not improve flow.

A comparative NHS command-center study found no significant, consistent improvement in patient-flow outcomes.



COMMAND CENTER



**Information without ownership and action rights is only
a better view of the bottleneck.**

Autonomy is a risk budget—not a feature.

Authority should narrow as consequence rises.

CONSEQUENCE RISES →



AUTOMATIC AUTHORITY NARROWS →

AUDIT · ROLLBACK · APPROVAL · STOP CONDITIONS · HUMAN ACCOUNTABILITY

Prove one flow in 90 days.

Buy fewer pilots. Create one closed-loop operating proof.

DAY 00-30

FRAME

One outcome
One owner
One baseline

DAY 31-60

SHADOW

Connect + replay
Test gates
Refine policy

DAY 61-90

TRANSACT

Reversible actions
Named approvals
Causal evaluation

ONE CAUSAL SCORECARD

ACCESS

FLOW

WORKFORCE

ECONOMICS

TRUST

SCALE ONLY WHEN THE SCORECARD MOVES

The future hospital is coordinated—not autonomous.

The winners will turn signals into safe action faster—
not simply accumulate more AI.

RUN THE TWIN →

01

ONE FLOW

02

ONE OWNER

03

ONE SCORECARD

